

Arsalan Esmaeili

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EDUCATION

University of Washington <i>PhD in Transportation Engineering</i>	(2023 – current)
Georgia Institute of Technology <i>MS in Computer Science</i>	(2025 – current)
University of Washington <i>MS in Industrial and Systems Engineering</i>	(2024 – 2026)
University of Tehran <i>MS in Transportation Engineering</i>	(2016 – 2019)
University of Tehran <i>BS in Civil Engineering</i>	(2011 – 2015)

RESEARCH EXPERIENCE

University of Washington	(2023 – current)
• AV Safety Pipelines: Developed computer vision and natural language processing pipelines to extract structured insights from unstructured autonomous vehicle crash reports, enabling scalable safety and risk analysis for next generation mobility. • Optimization Systems: Designed and implemented <i>Where2Charge</i>, a Python based decision support system that automates electric vehicle charging location recommendations by modeling complex spatial, network, and demand constraints. • Behavioral Modeling (Award-Winning): Led an empirical study on autonomous delivery vehicle acceptance using advanced behavioral modeling; recognized with the Best Paper Award (Top < 1%) at the 104th Transportation Research Board Annual Meeting. • Technical Curriculum Development: Contributed to the design and technical development of graduate-level “Autonomous Vehicles” course materials, translating research insights into deployable analytical workflows. • Logistics Digitization: Supported data modeling for a large scale urban freight digitization project, informing evidence-based strategies for curb access and utilization in last-mile delivery systems. • Equity Frameworks: Developed analytical frameworks focusing on equity and accessibility to evaluate electric vehicle infrastructure placement and quantify regional policy impacts. • Production Visualizations: Built interactive, production ready visualization dashboards using Streamlit to communicate high-velocity traffic crash patterns in Seattle, Washington, supporting data driven decision-making. • Big Data Analytics: Analyzed large-scale location-based service and GPS datasets to quantify accessibility to critical facilities during natural disasters, identifying infrastructure vulnerabilities. • Transit System Modeling: Processed and modeled transit operational datasets (automated passenger counter and smart card data) to characterize rider behavior patterns and support system optimization.	
University of Tehran	(2015 – 2019)
• Pedestrian Safety Modeling: Developed statistical models (including latent class cluster analysis and logit) to study factors associated with pedestrian crashes in collaboration with RMIT University, Australia. • National Data Strategy: Conducted feasibility studies for developing a national database for collecting and processing traffic crash data. • Behavioral Instrument Design: Developed a pedestrian behavior questionnaire to investigate safety-related behaviors in collaboration with Monash University, Australia. • Accessibility Engineering: Evaluated walkability and infrastructure barriers affecting pedestrians with physical disabilities. • Technical Communications: Served as an executive member and graphic designer for the 3rd and 5th International Traffic Accident Conference (ITAC), University of Tehran.	

LEADERSHIP AND PROFESSIONAL EXPERIENCE

University of Washington	(2024 – current)
• Lab Manager, Urban Freight Lab (UFL), supporting lab operations, research coordination.	

- Instructor, graduate-level Travel Demand Forecasting course, providing hands-on instruction in Python-based data analysis and travel demand modeling.
- Lab Session Lead, guiding students through applied analytical workflows, including data preparation, model estimation, and results interpretation.
- Guest Lecturer and Teaching Assistant, Supply Chain, Transportation, and Logistics master’s program, supporting applied analytics and decision-oriented coursework.
- Course Development Contributor, Autonomous Vehicles course, supporting course design and development

University of Tehran

(2016 – 2019)

- Lecturer and Project Developer, Road Geometric Design Project course.
- Elected Student Representative, College of Engineering.
- Elected Executive Member, Students’ Scientific Association, College of Engineering.
- Executive Committee Member and Poster Designer, Annual Alumni Association Convention, Faculty of Engineering.
- Design Team Lead, commemoration statuette for the Alumni Association of the University of Tehran.
- Executive Staff Member and Poster Designer, 3rd and 5th International Traffic Accident Conferences.

Professional Projects and Applied Policy Work

(2017 – 2022)

- **Project manager, global road safety policy design (WHO):** Collaborated with the World Health Organization on national road safety strategic frameworks and speed management guidelines for rural road networks.
- **Research and development engineer, highway safety infrastructure:** Analyzed operational and geometric design alternatives for median openings and U-turn management in highway safety projects.
- **Project manager, municipal safety strategy:** Led data-driven safety improvement planning for a large metropolitan transportation agency.

TECHNICAL SKILLS

Statistical, Machine Learning, and Behavioral Modeling: Markov decision processes and reinforcement learning; Bayesian modeling; structural equation modeling and factor analysis; discrete choice experiments and models, including mixed logit; latent class analysis; clustering and dimensionality reduction methods; time series modeling, including Bayesian structural time series.

Large-Scale Mobility Data Engineering and Analytics: Processing and analysis of large-scale spatiotemporal data, including GPS, location-based service data, transit smart card data, and automated passenger counter data; survey design, data integration, and analysis; data quality assessment and feature engineering.

Computer Vision, NLP, and Unstructured Data Analysis: Image processing, convolutional neural networks, and computer vision methods for safety and infrastructure analysis; natural language processing for structuring and analyzing unstructured textual data, including crash reports.

Geospatial Analysis, Simulation, and Transportation Systems Modeling: Spatial data analysis; accessibility modeling; traffic simulation; infrastructure planning and evaluation; ArcGIS; Civil 3D; AutoTURN; Vissim; Synchro.

Optimization and Decision-Support Systems: Spatial optimization; dynamic programming; mixed-integer programming; development of decision-support tools for infrastructure planning and policy evaluation.

Programming and Technical Computing: Python (primary); SQL; R; JavaScript; MATLAB.

Analytical, Survey, Visualization, and Workflow Tools: Stata; SPSS; NLOGIT; Streamlit; Excel; Qualtrics; Git/GitHub.

SELECTED PUBLICATIONS

A. Esmaili, S. Keshavarz, M. Oshanreh, and A. Regan, “Improving Autonomous Vehicle Crash Data by Data Mining and Artificial Intelligence Tools,” *Transportation Research Board*

A. Esmaili, K. Aghabayk, S. Rejali, A. Mohammadi, and C. De Gruyter, “Acceptance of Autonomous Delivery Vehicles in an Eastern Culture: The Moderating Role of Perceived Security Risk of Theft,” *Transport Policy*

A. Esmaili, M. Oshanreh, S. Naderian, D. MacKenzie, and C. Chen, “Assessing the Spatial Distributions of Public Electric Vehicle Charging Stations with Emphasis on Equity Considerations in King County, Washington,” *Sustainable Cities and Society*

A. Esmaili, K. Aghabayk, N. Parishad, and A. N. Stephens, “Investigating the Interaction between Pedestrian Behaviors and Crashes through Validation of a Pedestrian Behavior Questionnaire (PBQ),” *Accident Analysis & Prevention*.

A. Esmaili, K. Aghabayk, and N. Shiwakoti, “Latent Class Cluster Analysis and Mixed Logit Model to Investigate Pedestrian Crash Injury Severity,” *Sustainability*